

Implementation Plan: Documentation Agent Pipeline

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Artifact type: plan

Branch: `001-documentation-agent` | **Date**: 2026-07-16 | **Spec**: spec.md

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Implementation Plan: Documentation Agent Pipeline

Branch: `001-documentation-agent` | **Date**: 2026-07-16 | **Spec**: spec.md

Input: Feature specification from `/specs/001-documentation-agent/spec.md`

Summary

Build a Spec Kit extension and backend pipeline that discovers markdown artifacts from the supported spec directories, classifies them by role, deduplicates unchanged content, validates structured output, routes content through an explicit agent-style transformation stage, renders PDF documents deterministically, and persists versioned results in PostgreSQL. The implementation will include the extension command set, the FastAPI endpoints, the PostgreSQL schema, the transformation service, and the rendering pipeline defined in the feature spec, while keeping frontend work out of scope.

Technical Context

Language/Version: Python 3.11+ **Primary Dependencies**: FastAPI, PostgreSQL, WeasyPrint, Spec Kit extension framework, pytest **Storage**: PostgreSQL with the exact

three-table schema from the feature spec: projects, artifacts, and doc_versions ****Testing****: pytest for backend unit and integration tests ****Target Platform****: Linux-based local development environment with Docker support ****Project Type****: Web service plus Spec Kit extension package ****Performance Goals****: Support the documented ingestion and rendering workflow for repository-spec artifacts without introducing unnecessary complexity or extra services ****Constraints****: The implementation must preserve the agent-native path, the hook-fallback path, the exact API routes, the exact extension command set, and the non-goal boundaries from the feature spec. Artifact type must be handled in application code rather than as a database-enforced CHECK enum. ****Scale/Scope****: Single repository, single project context, small extension and backend footprint

Constitution Check

GATE: Must pass before Phase 0 research. Re-check after Phase 1 design. - The plan stays within the Spec Kit-side extension, backend/API, pipeline, and Docker/Postgres scope for this feature. - The plan preserves the zero-config agent-native path; no separate AI provider configuration is added for the primary flow. - The plan does not introduce frontend work, ticket-board/Kanban functionality, or other non-goal scope. - The plan respects the explicit non-goal boundaries from the feature spec.

Project Structure

Documentation (this feature)

```
``text specs/001-documentation-agent/ ■■■■ plan.md ■■■■ research.md ■■■■
data-model.md ■■■■ quickstart.md ■■■■ contracts/ ■■■■ tasks.md ```
```

Source Code (repository root)

```
``text extension/ ■■■■ extension.yml ■■■■ commands/ ■ ■■■■ setup.md ■ ■■■■ docgen.md
■ ■■■■ status.md ■ ■■■■ regenerate.md ■■■■ scripts/ ■ ■■■■ python/ ■ ■■■■
post_commit_hook.py ■■■■ config-template.yml ■■■■ README.md ■■■■ LICENSE backend/
■■■■ app/ ■ ■■■■ api/ ■ ■ ■■■■ routes.py ■ ■ ■■■■ deps.py ■ ■■■■ services/ ■ ■ ■■■■
ingestion.py ■ ■ ■■■■ validation.py ■ ■ ■■■■ rendering.py ■ ■ ■■■■ persistence.py ■ ■■■■
models/ ■ ■ ■■■■ schemas.py ■ ■■■■ repositories/ ■ ■ ■■■■ artifact_repo.py ■ ■■■■
main.py ■■■■ tests/ ■ ■■■■ contract/ ■ ■■■■ integration/ ■ ■■■■ unit/ ■■■■ requirements.txt
infra/ ■■■■ docker-compose.yml ``` **Structure Decision**: The implementation will be split
into a Spec Kit extension package plus a FastAPI backend service, with an explicit agent-style
transformation service handling title/abstract/section planning before persistence and
rendering, and with a richer document renderer that builds a polished cover page, table of
contents, grouped sections, and footer metadata for the PDF output.
```

Complexity Tracking

No constitution violations are required for this plan.

source: specs/001-documentation-agent/plan.md | commit: n/a